

Minimizing Required Data in a Neural Network Accelerator Using Double-Stage Weight Sharing

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- **0. Background**
- **1-1. Double-Stage Weight sharing**
- **1-2. Design Exploration**
- **2. Experiment**
- **3. Conclusion**

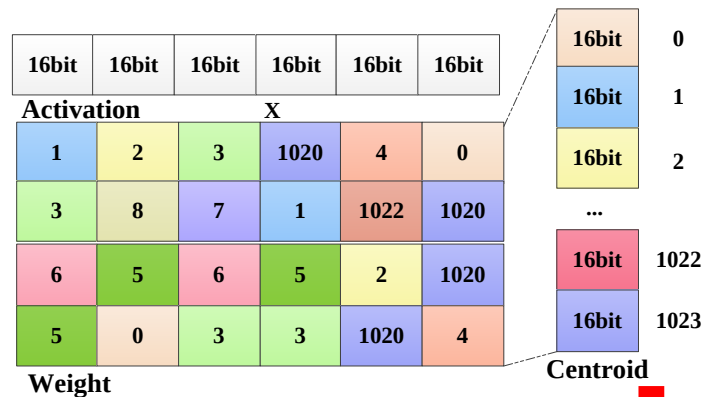
- **The demand for FC layer increases**
 - ex) *Self-Attention (ICLR'15)*
 - Too many parameters without reuse
- **Problem of previous accelerators**
 - “The **on-chip buffers** consume significant chip area(up to 70%)...”
 - *TETRIS(OSR'17)*
 - “The area(93%) and power(59%) is dominated by **SRAM**”
 - *EIE(ISCA'16)*
- **Objective**
 - Put all parameters in on-chip memory
 - Minimize SRAM by reducing required data
 - Double-stage weight sharing
 - Including Hashing trick (ICML'16)

1-1. Double-Stage Weight sharing

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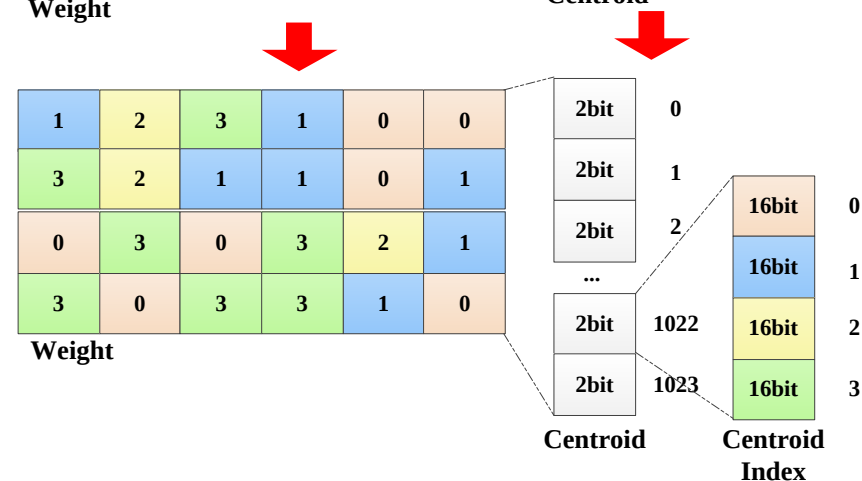
- **Original Hashing trick**

- Statically shared mapping
 - Hard to express with small centroid
 - ex) 98000 centroids for 784000



- **Resolution**

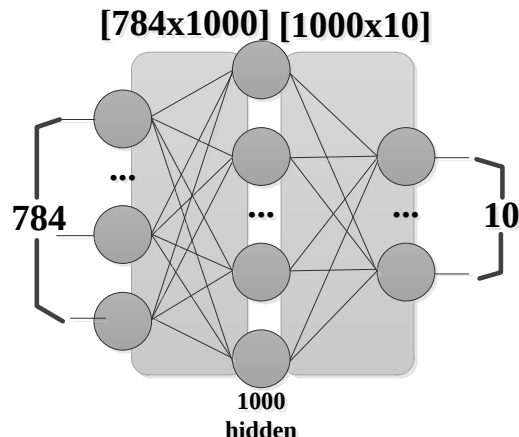
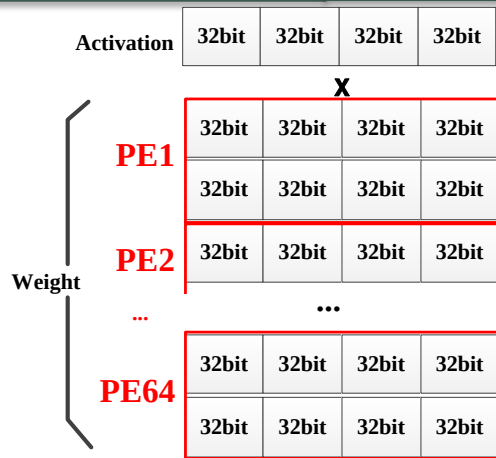
- Two stage weight sharing
 - 1. Scatter centroids with hashing trick
 - 2. Cluster them with index



1-1. Double-Stage Weight sharing

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- **For Hardware : Blocked Sharing**
 - Reduce Random Access – *BHNN (DAC'17)*
- **Training Step**
 - Step1. Pretrain
 - Pretrain with normal training
 - Step2. Training : Hashing trick
 - Initialization : same index $\rightarrow$ mean value
 - Apply gradients : same index $\rightarrow$ mean gradients
 - Step3. Post training : with index
 - Initialization : **k-means clustering** of centroids
 - Apply gradients : same index $\rightarrow$ mean gradients



1-1. Double-Stage

Centroid
Index

2bit	2bit	2bit	2bit
0	1	2	3

...

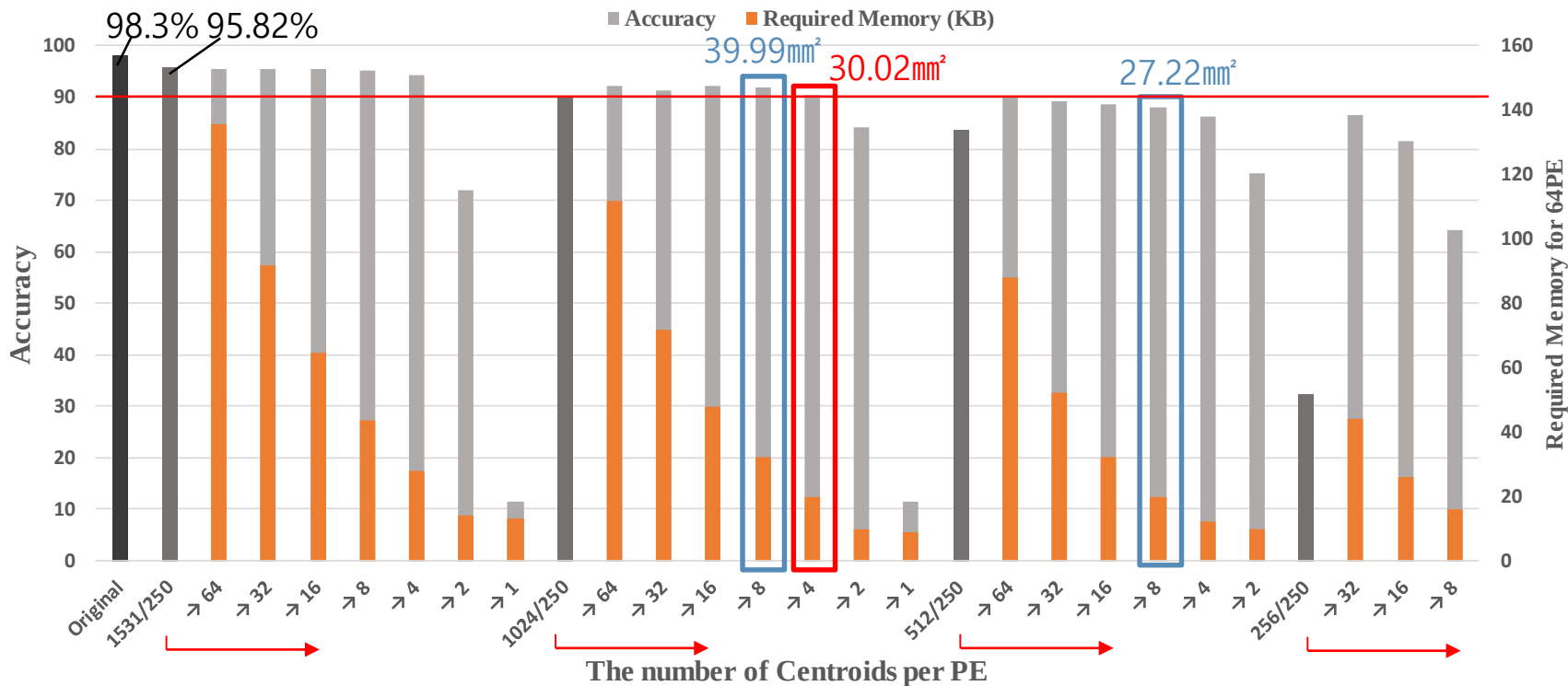
2bit	2bit	2bit
1021	1022	1023

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Centroid

16bit	16bit	16bit	16bit
0	1	2	3

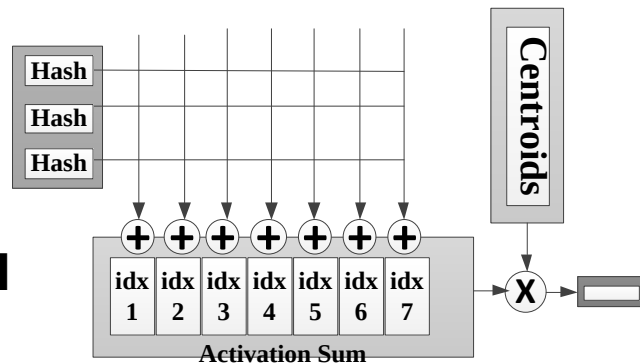
- The # of Centroids



1-2. Design Exploration

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- **Design Exploration for PE**
 - 1. PARALLEL
 - LUT problem for weight sharing
 - Suggested **post-multiplication method**
 - 2. SKIP



	PARALLEL(A+W)		SKIP(A+W)		Post-multiply	Disadvantage
	A	W	A	W		
D1	Serial	Serial	skip	skip(CSC)	Impossible	Hashing Trick has too many parameters for centroids (Impossible)
D2	Serial	Parallel($\updownarrow$)	skip	skip(CSC)	Impossible	Register Share Problem (Impossible)
D3			skip	X	Impossible	LUT problem (SRAM size $\uparrow$)
D4	Parallel($\leftrightarrow$)	Parallel($\leftrightarrow$)	skip	X	Possible	LUT problem $\rightarrow$ solved
D5			X	skip(CSR)	Possible	No Area optimization : Meaningless

2. Experiment

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- **Implemented with Verilog**
 - Synthesized with *Design Complier* under 32nm CMOS tech
 - Area was estimated with *Synopsys IC Compiler*
 - Compared with EIE
- **Performance**
 - Sparsity : W(0.9), ACT(0.7)
 - 37.5% cycles ↓ = **60%** performance
- **SRAM size / Area**
 - SRAM size : 10,384KB→16KB
 - Area : **26.48%** area ↓
- **Performance-area efficiency**
 - **60% ↑ / 26.48% ↓ = 117.63% ↑**

			EIE (ISCA'16)	Ours
Performance Ratio (cycle)			468.75u	292.97u
Mem	Register size		2KB	16KB
	SRAM	Size	10,368 KB	16KB
		Area	38.83 mm ²	0.76 mm ²
64PE	Logic gate count		(40,251)	20,457,792
	Area	Logic	2.00 mm ²	29.26 mm ²
		Total	40.83 mm ²	30.02 mm ²

3. Conclusion

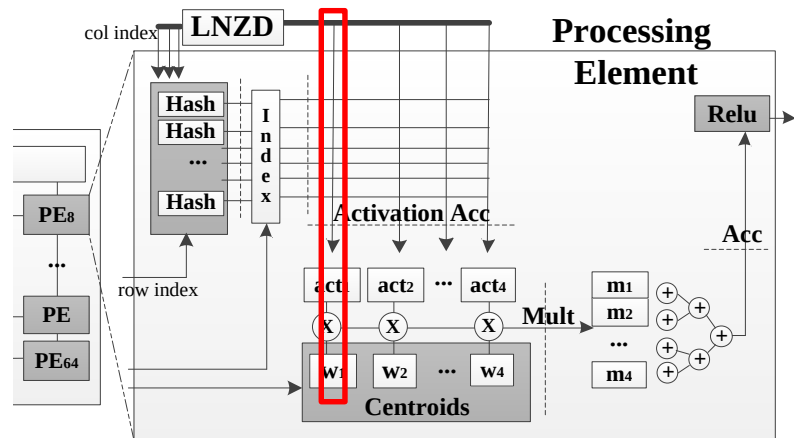
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• Contribution

- Minimized required data with 2-stage weight sharing and design exploration
- Improved area-performance efficiency

• Future Work

- Power
 - Amdahl's law $\rightarrow 22.36\% \uparrow$
 - 1. Pruning : 86.97% (1024 $\rightarrow$ 3/4)
 - 29.54mm²
 - 2. Change XXhash : *BHNN*



Hashed : CCU + 64 PE			
CCU	PE : 16 XXhash + Act_sum + Register		
	XXhash	Act_sum	Regs
0.01%	41.89%	3.4%	53.55%

<The proportions of gates>

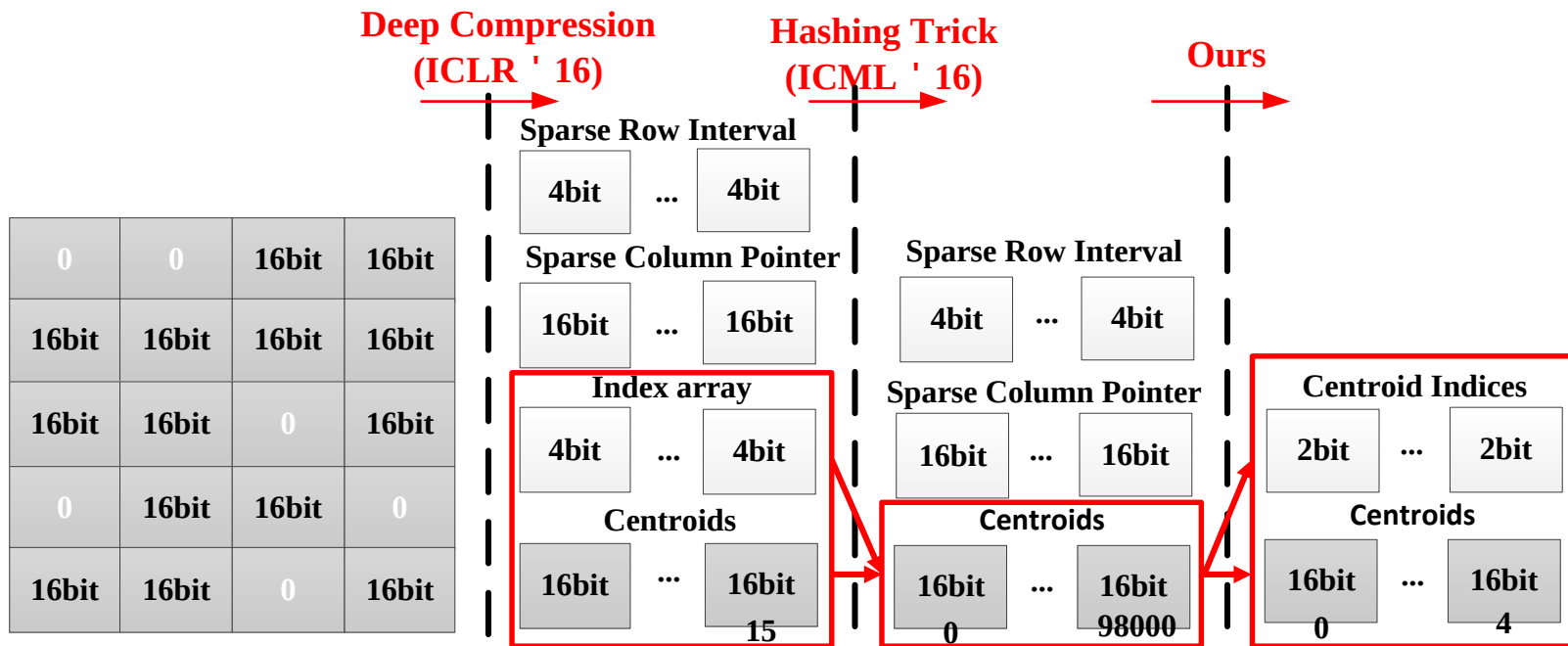


Appendix1. Data format change

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- **Result of required data**

- Double-stage weight sharing + No Sparsity + Parallelization



Appendix2. Design factors

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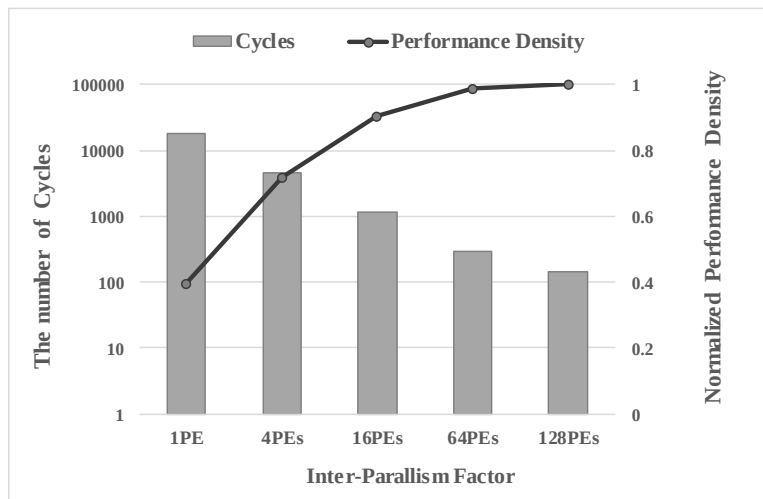
- **2. Intra-PE parallelism**

- DRAM bandwidth
- Snapdragon 820,835, 845, 850 (29.8GB/s)
 - Bandwidth : 29.8 GB/s
 - Core 800MHz = 1.25ns/cycle
 - 1cycle : 298 bits > 16bits x **16**

- **3. Inter-PE parallelism**

- Performance / Area for PEs
 - Seems to converge
- Used **64PEs** to compare with EIE

	SKIP					
Activation	32bit	32bit	32bit	32bit	32bit	32bit
	X					
	0	0	1	2	3	1
	1	1	3	2	1	1
	1	1	0	1	0	1
Weight	0	1	1	0	3	3



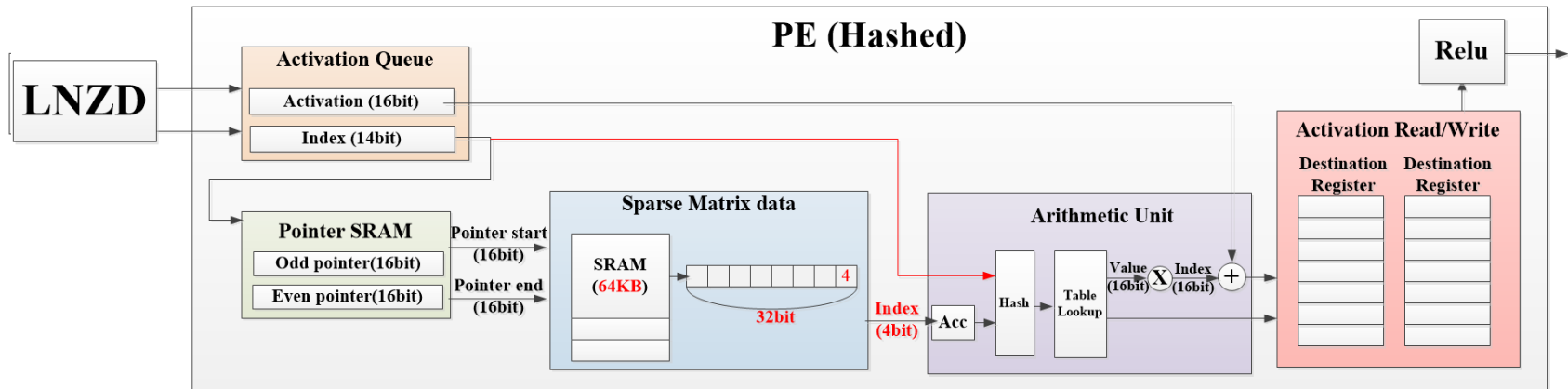
- **Simplest design**

- **EIE SRAM**

- 10752KB : 2048KB(Ptr) + 8192KB(SPmat) + 128KB(Act) + LUT(512KB)

- **Hashing trick SRAM**

- 6784KB(36.9%↓) : 2048KB(Ptr) + 4096KB + 128KB(Act) + LUT(512KB)
 - Area not much influenced : **Hash unit**

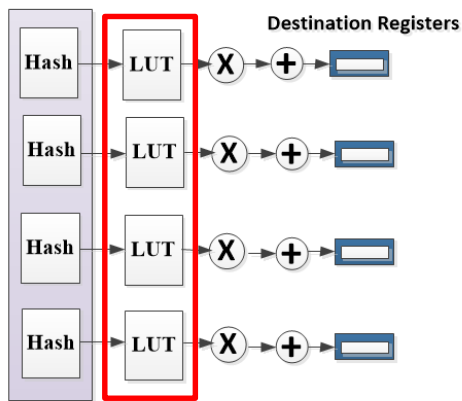


Appendix4. LUT problem

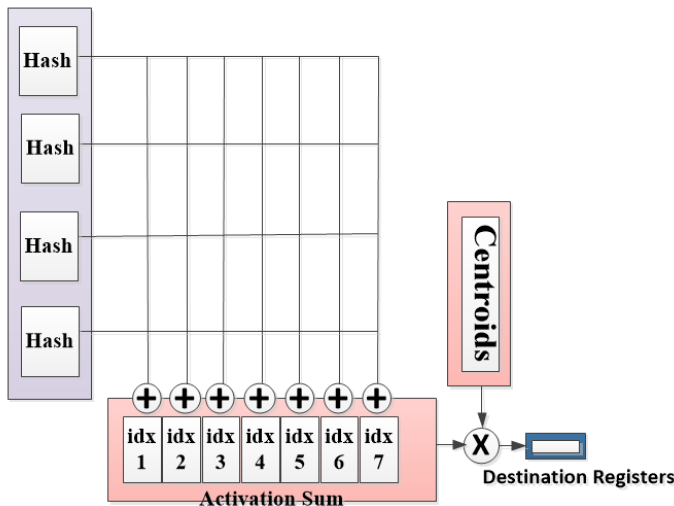
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- **Design Exploration for PE**
 - 1. PARALLEL : Weight sharing problem
 - Many Look Up Tables (SRAM size $\uparrow$)
 - Suggested **post-multiplication method**

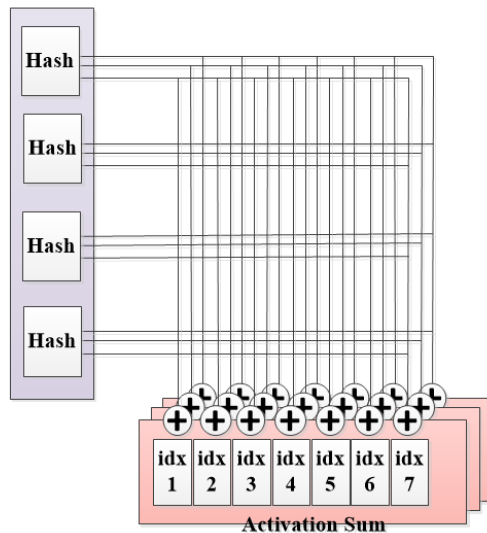
<Parallel without post-mult>



<one output : Possible>



<three outputs : Impossible>



Appendix4. Future work

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• Future Work

- Size of workload
 - Ours : 1M parameters
 - VGG : 124M parameters
- Power
 - Amdahl's law $\rightarrow 22.36\% \uparrow$
 - 1. Pruning : 86.97% (1024 $\rightarrow$ 4 cen)
 - 2. Change XXhash : *BHNN (DAC'17)*
- Conv layer
 - Winograd : 1x1, 3x3 filter
 - Reuse : efficient?

<The proportions of gates>

Hashed : CCU + 64 PE			
CCU	PE : 16 XXhash + Act_sum + Register		
	XXhash	Act_sum	Regs
0.01%	41.89%	3.4%	53.55%

